

Course 5303B

5 Credits: 4

Program Elective

Source-grounded

Non-Relational Database

□ Basics of database is studied already

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5303B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5303B.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	5303B
Course title	Non-Relational Database
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

29 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Basics of database is studied already
- To develop basic concepts and appreciate the applications of database systems. Become familiar with the expanding role of database technology.
- systems. Understand the basic architecture and data models of a NoSQL
- To understand Emerging features in the area of database security
- To demonstrate the basic concepts of transaction processing and concurrency

Course outcomes

CO	Official outcome	Study level
CO1	Databases Demonstrate the features of indexing and 17 Understand	See official syllabus
CO2	hashing in database applications To Demonstrate the basic concepts of 15 Understand	See official syllabus
CO3	transaction processing and concurrency control Identify security and integrity threads in 14 Understanding	See official syllabus
CO4	database	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 3 3 2
CO2	CO2 3 2 3 3 3
CO3	CO3 3 2 3 3 2

Row	Extracted official mapping
CO4	CO4 3 3 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain and compare different types of NoSQL Databases	8	As specified
Module 2	Demonstrate the features of indexing and hashing in database applications	7	As specified
Module 3	concurrency control	7	As specified
Module 4	Identify security and integrity threads in database	7	As specified

MODULE 1

Explain and compare different types of NoSQL Databases

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain and compare different types of NoSQL Databases
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key

NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, nosql data model: aggregate models- 2 understanding document data model- key should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-നോSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key ഡോക്യുമെന്റ് ഡാറ്റാ മോഡൽ definition/meaning ഡോക്യുമെന്റ് ഡാറ്റാ മോഡൽ, steps, application ഡോക്യുമെന്റ് ഡാറ്റാ മോഡൽ. Technical terms English-നോSQL ഡാറ്റാ മോഡൽ.

Key-Value Data Model Columnar Data Model

Key-Value Data Model Columnar Data Model is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Key-Value Data Model Columnar Data Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, key-value data model columnar data model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Key-Value Data Model Columnar Data Model means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-☐☐ Key-Value Data Model Columnar Data Model
☐☐☐☐☐☐☐☐☐ ☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐ ☐☐☐☐☐☐,
steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐
☐☐☐☐☐☐☐☐☐☐.

Graph Based Data Model

Graph Based Data Model is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Graph Based Data Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, graph based data model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Graph Based Data Model means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഗ്രാഫ് Based Data Model ന്റെ നിർവ്വചനം/അർത്ഥം, steps, application ന്റെ വിവരണം. Technical terms English-ൽ ഉൾപ്പെടുത്തിയിരിക്കുന്നു.

NoSQL system ways to handle big data 2 Understanding problems

NoSQL system ways to handle big data 2 Understanding problems is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify NoSQL system ways to handle big data 2 Understanding problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, nosql system ways to handle big data 2 understanding problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What NoSQL system ways to handle big data 2 Understanding problems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- NoSQL system ways to handle big data 2
Understanding problems definition/meaning .
 , steps, application . Technical
terms English- .

Moving Queries to data, not data to the query

Moving Queries to data, not data to the query is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Moving Queries to data, not data to the query using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, moving queries to data, not data to the query should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Moving Queries to data, not data to the query means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്നിരിക്കുന്ന ചോദ്യങ്ങൾക്ക് ഉത്തരം നൽകുക. ചോദ്യങ്ങൾക്ക് definition/meaning, steps, application എന്നിവ ഉൾപ്പെടെയുള്ള ഉത്തരങ്ങൾ നൽകുക. Technical terms English-ൽ ഉൾപ്പെടെ ഉത്തരങ്ങൾ നൽകുക.

Hash rings to distribute the data on clusters

Hash rings to distribute the data on clusters is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Hash rings to distribute the data on clusters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, hash rings to distribute the data on clusters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Hash rings to distribute the data on clusters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Hash rings to distribute the data on clusters
definition/meaning .
steps, application . Technical terms English-
.

Replication to scale reads

Replication to scale reads is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Replication to scale reads using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, replication to scale reads should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Replication to scale reads means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Replication to scale reads $\frac{1}{2}$ definition/meaning $\frac{1}{2}$. $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$, steps, application $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$. Technical terms English- $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$.

Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes.

Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, database distributed queries to data nodes. 2 understanding nosql data model: aggregate models- document data model- key-value data model columnar data model, graph based data model,, nosql system ways to handle big data problems, moving queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, database distributed queries to data nodes.

should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Explain and compare different types of NoSQL Databases

M1.01	NoSQL Data model: Aggregate Models- Understanding Document Data Model- Key	2
M1.02	Key-Value Data Model Columnar Data Model Understanding	2
M1.03	Graph Based Data Model Understanding	1
M1.04	NoSQL system ways to handle big data Understanding problems	2
M1.05	Moving Queries to data, not data to the query Understanding	1
M1.06	Hash rings to distribute the data on clusters Understanding	1
M1.07	Replication to scale reads Understanding	1
M1.08	Database distributed queries to Data nodes. Understanding	2

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Demonstrate the features of indexing and hashing in database applications

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate the features of indexing and hashing in database applications
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Data Storage and Indexes- File organizations 2 Understanding.

Data Storage and Indexes- File organizations 2 Understanding. is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Storage and Indexes- File organizations 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data storage and indexes- file organizations 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Data Storage and Indexes- File organizations 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- Data Storage and Indexes- File organizations 2 Understanding. definition/meaning . steps, application . Technical terms English- .

Primary indexing

Primary indexing is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Primary indexing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, primary indexing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Primary indexing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Secondary index structures

Secondary index structures is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Secondary index structures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, secondary index structures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Secondary index structures means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Secondary index structures
definition/meaning
application
Technical terms English-
.

Various index structures – hash-based

Various index structures – hash-based is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Various index structures – hash-based using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, various index structures – hash-based should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Various index structures – hash-based means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Various index structures – hash-based
index structures definition/meaning . Steps, application . Technical terms English-
index structures.

Dynamic hashing techniques

Dynamic hashing techniques is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Dynamic hashing techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, dynamic hashing techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Dynamic hashing techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- dynamic hashing techniques
 definition/meaning .
 application
 Technical terms English-
 .

Multi-level indexes,

Multi-level indexes, is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Multi-level indexes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, multi-level indexes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Multi-level indexes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Multi-level indexes, definition/meaning . steps, application . Technical terms English- .

B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and

B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, b and b+ trees(insertion and deletion) 2 applying data storage & indexing data storage and indexes: data storage and indexes - file organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, b and b+ trees(insertion and deletion) to demonstrate the basic concepts of transaction processing and should be discussed with

attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and definition/meaning. , steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Demonstrate the features of indexing and hashing in database applications

M2.01	Data Storage and Indexes- File organizations Understanding.	2
M2.02	Primary indexing Understanding	3
M2.03	Secondary index structures Understanding	2
M2.04	Various index structures – hash-based Applying	3
M2.05	Dynamic hashing techniques Applying	3
M2.06	Multi-level indexes, Applying	2
M2.07	B and B+ trees(insertion and deletion) Applying	2
	Series Test – I	1

Contents:

Data Storage & Indexing Data Storage and Indexes:

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

concurrency control

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: concurrency control
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Transaction concept, transaction state

Transaction concept, transaction state is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Transaction concept, transaction state using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transaction concept, transaction state should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Transaction concept, transaction state means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-Transaction concept, transaction state
Transaction concept definition/meaning Transaction state definition, steps, application Transaction state definition. Technical terms English-Transaction state
Transaction state.

Implementation of atomicity and durability

Implementation of atomicity and durability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implementation of atomicity and durability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implementation of atomicity and durability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implementation of atomicity and durability means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓം Implementation of atomicity and durability
ഓംഓംഓംഓംഓം ഓംഓം definition/meaning ഓംഓംഓംഓംഓംഓം. ഓംഓംഓം ഓംഓംഓം ഓംഓംഓം,
steps, application ഓംഓംഓം ഓംഓംഓംഓം ഓംഓംഓം. Technical terms English-ഓം ഓംഓംഓം
ഓംഓംഓംഓംഓം.

Concurrent executions

Concurrent executions is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Concurrent executions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, concurrent executions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Concurrent executions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Concurrent executions
 definition/meaning .
 , steps, application
 . Technical terms English-
 .

Serializability

Serializability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Serializability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, serializability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Serializability means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-Serializability ഓരോന്നിനും ഓരോന്നിനും definition/meaning ഓരോന്നിനും. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഓരോന്നിനും ഓരോന്നിനും. Technical terms English-ഓരോന്നിനും ഓരോന്നിനും.

Recoverability

Recoverability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recoverability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recoverability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recoverability means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി
definition/meaning റീക്കവറബിലിറ്റി. റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി, steps, application
റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി. Technical terms English-റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി.

Implementation of isolation

Implementation of isolation is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implementation of isolation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implementation of isolation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implementation of isolation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Implementation of isolation
definition/meaning .
application . Technical terms English-
.

Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability.

Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, testing for serializability 2 understanding transactions management: transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, serializability, recoverability, implementation of isolation, transaction definition in sql, testing for serializability. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. definition/meaning .
, steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

concurrency control

M3.01	Transaction concept, transaction state	2
Understanding		
M3.02	Implementation of atomicity and durability	2
Understanding		
M3.03	Concurrent executions	2
Applying		
M3.04	Serializability	3
Applying		
M3.05	Recoverability	2
Understanding		
M3.06	Implementation of isolation	2
Understanding		
M3.07	Testing for Serializability	2
Understanding		

Contents:

TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Identify security and integrity threads in database

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify security and integrity threads in database
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Security and integrity threats, Defence 2 Understanding mechanisms

Security and integrity threats, Defence 2 Understanding mechanisms is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security and integrity threats, Defence 2 Understanding mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security and integrity threats, defence 2 understanding mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Security and integrity threats, Defence 2 Understanding mechanisms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- സുരക്ഷയും അഭിമാനത്തിന്റെ ഭംഗവും, Defence 2

Understanding mechanisms സുരക്ഷയുടെ പ്രവർത്തനം definition/meaning

സുരക്ഷയുടെ പ്രവർത്തനം. സുരക്ഷയുടെ പ്രവർത്തനം, steps, application സുരക്ഷയുടെ പ്രവർത്തനം

സുരക്ഷയുടെ പ്രവർത്തനം. Technical terms English- സുരക്ഷയുടെ പ്രവർത്തനം.

Statistical database auditing & control

Statistical database auditing & control is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Statistical database auditing & control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, statistical database auditing & control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Statistical database auditing & control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Statistical database auditing & control

Statistical database auditing & control definition/meaning. Steps, application. Technical terms English-Malayalam.

Security issue based on granting/revoking of 2 Understanding privileges

Security issue based on granting/revoking of 2 Understanding privileges is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security issue based on granting/revoking of 2 Understanding privileges using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security issue based on granting/revoking of 2 understanding privileges should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Security issue based on granting/revoking of 2 Understanding privileges means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- സുരക്ഷാ പ്രശ്നം based on granting/revoking of 2
Understanding privileges സുരക്ഷാ പ്രശ്നം definition/meaning സുരക്ഷാ പ്രശ്നം.
സുരക്ഷാ പ്രശ്നം സുരക്ഷാ പ്രശ്നം, steps, application സുരക്ഷാ പ്രശ്നം സുരക്ഷാ പ്രശ്നം. Technical
terms English- സുരക്ഷാ പ്രശ്നം സുരക്ഷാ പ്രശ്നം.

Introduction to statistical database security

Introduction to statistical database security is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction to statistical database security using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction to statistical database security should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction to statistical database security means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇം Introduction to statistical database security
ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms English-ഇം ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

PL/SQL Security - Locks

PL/SQL Security - Locks is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PL/SQL Security - Locks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pl/sql security - locks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PL/SQL Security - Locks means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-PL/SQL Security - Locks ഫലപ്രസാദം ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം ഫലപ്രസാദം.

Implicit locking, types and levels of locks, 2 Understanding explicit locking

Implicit locking, types and levels of locks, 2 Understanding explicit locking is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implicit locking, types and levels of locks, 2 Understanding explicit locking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implicit locking, types and levels of locks, 2 understanding explicit locking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implicit locking, types and levels of locks, 2 Understanding explicit locking means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-
Understanding explicit locking
definition/meaning
steps, application
Technical terms English-

Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security - Locks - Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html

Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security - Locks - Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources Sl. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, oracles' named exception handlers 2 understanding database security security and integrity threats, defence mechanisms, statistical database auditing & control. security issue based on granting/revoking of privileges, introduction to statistical database security. pl/sql security – locks – implicit locking, types and levels of locks, explicit locking, oracles' named exception handlers. text / reference t/r book title/author elmasri r. and s. navathe, database systems: models, languages, design r1 and application programming, pearson education, 2013. r2 adam fowler, nosql for dummies, john wiley & sons, 2015 nosql data models: trends and challenges (computer engineering: databases r3 and big data), wiley, 2018 handbook of database security, applications and trends r4 gertz, jajodia online resources sl. no website link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue

Aspect	What to prepare
	based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles’ named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 https://www.w3resource.com/redis/ 2 https://www.w3schools.in/category/mongodb/ 3 https://www.tutorialspoint.com/arangodb/index.html 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Oracles’ named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles’ named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Identify security and integrity threads in database

M4.01	Security and integrity threats, Defence	2
Understanding	mechanisms	
M4.02	Statistical database auditing & control	2
Understanding		
M4.03	Security issue based on granting/revoking of	2
Understanding	privileges	
M4.04	Introduction to statistical database security	2
Understanding		
M4.05	PL/SQL Security – Locks	2
Understanding		
M4.06	Implicit locking, types and levels of locks,	2
Understanding	explicit locking	
M4.07	Oracles' named Exception Handlers	2

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key. (3 marks)

Answer: Begin with the standard definition or identity of *NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Key-Value Data Model Columnar Data Model. (3 marks)

Answer: Begin with the standard definition or identity of *Key-Value Data Model* *Columnar Data Model*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Graph Based Data Model. (3 marks)

Answer: Begin with the standard definition or identity of *Graph Based Data Model*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain NoSQL system ways to handle big data 2 Understanding problems. (3 marks)

Answer: Begin with the standard definition or identity of *NoSQL system ways to handle big data 2 Understanding problems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Data Storage and Indexes- File organizations 2 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Data Storage and Indexes- File organizations 2 Understanding..*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Primary indexing. (3 marks)

Answer: Begin with the standard definition or identity of *Primary indexing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Secondary index structures. (3 marks)

Answer: Begin with the standard definition or identity of *Secondary index structures*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Various index structures - hash-based. (3 marks)

Answer: Begin with the standard definition or identity of *Various index structures - hash-based*. State the governing principle, list the key components or stages, and close

with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 3

Define or explain Transaction concept, transaction state. (3 marks)

Answer: Begin with the standard definition or identity of *Transaction concept*, *transaction state*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Implementation of atomicity and durability. (3 marks)

Answer: Begin with the standard definition or identity of *Implementation of atomicity and durability*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Concurrent executions. (3 marks)

Answer: Begin with the standard definition or identity of *Concurrent executions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

□□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Serializability. (3 marks)

Answer: Begin with the standard definition or identity of *Serializability*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Module 4

Define or explain Security and integrity threats, Defence 2 Understanding mechanisms. (3 marks)

Answer: Begin with the standard definition or identity of *Security and integrity threats, Defence 2 Understanding mechanisms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Statistical database auditing & control. (3 marks)

Answer: Begin with the standard definition or identity of *Statistical database auditing & control*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

Define or explain Security issue based on granting/revoking of 2 Understanding privileges. (3 marks)

Answer: Begin with the standard definition or identity of *Security issue based on granting/revoking of 2 Understanding privileges*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Introduction to statistical database security. (3 marks)

Answer: Begin with the standard definition or identity of *Introduction to statistical database security*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Data Storage and Indexes- File organizations 2 Understanding..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Transaction concept, transaction state.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Security and integrity threats, Defence 2 Understanding mechanisms.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link <https://www.w3resource.com/redis/>
<https://www.w3schools.in/category/mongodb/>
- <https://www.tutorialspoint.com/arangodb/index.html>

- https://www.tutorialspoint.com/cassandra/cassandra_introduction.html

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5303B. Verify institutional updates against the current official curriculum.